

**Information
Visualization**

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Introduction to **RDFLib**

Lesson 3

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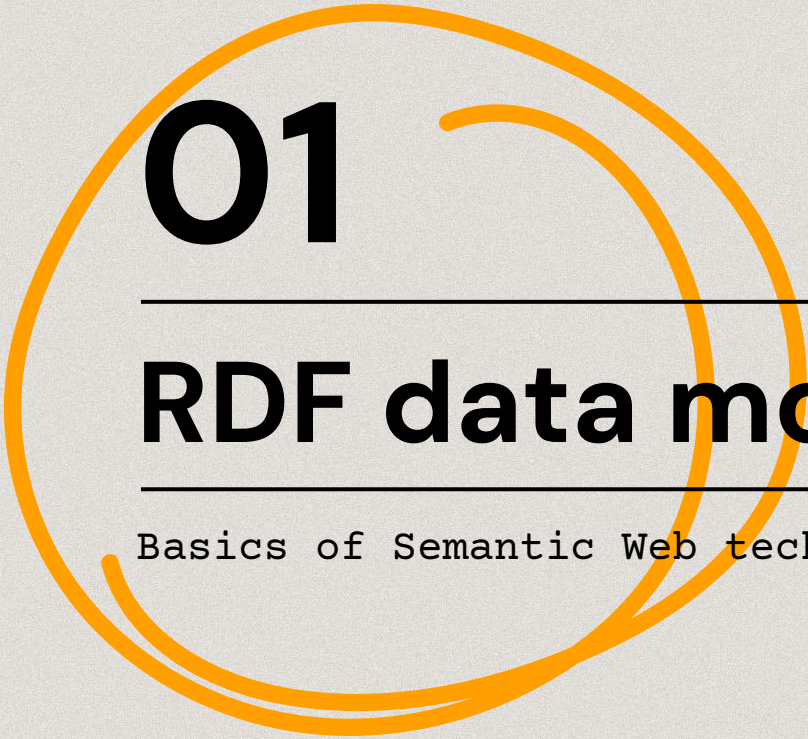
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serialisations, and named graphs

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with RDFLib



01

RDF data model

Basics of Semantic Web technologies

The web we know

01

URI



UNIFORM RESOURCE
IDENTIFIER

Identify the
location of
documents on the web

HTTP



HYPertext TRANSFER
PROTOCOL

It's the protocol
for exchanging data
and documents on the
web

HTML



HYPertext MARKUP
LANGUAGE

It's the markup
language for
documents returned
via HTTP

RDF

The web of data

01

URI



UNIFORM RESOURCE
IDENTIFIER

HTTP



HYPERTEXT TRANSFER
PROTOCOL

HTML



HYPERTEXT MARKUP
LANGUAGE

RDF

In the Web of Data a URI is a ***persistent conceptual mapping*** to a real entity (e.g. a person), not to a HTML page.

- If the **location** of a HTML file describing the entity changes, the URI does not change.
- Many HTML pages can describe the **same entity** identified by the same URI.
- A HTML page can include information about **many entities**, hence being linked to many URIs.

RDF

RDF

URI

subject

URI

predicate

URI

object

RESOURCE DESCRIPTION FRAMEWORK

In the WoD a URI identifies both **real entities** and the **relations** (links) between them.

Every piece of information is represented as a **triplet** of URIs, identifying respectively a subject, a predicate, and an object.

RDF example

https://en.wikipedia.org/wiki/Robert_Capa



WIKIPEDIA
The Free Encyclopedia

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Robert Capa

From Wikipedia, the free encyclopedia

Robert Capa (born **Endre Ernő Friedmann**;^[1] October 22, 1913 – May 25, 1954) was a Hungarian-American war photographer and photojournalist as well as the companion and professional partner of photographer **Gerda Taro**. He is considered by some to be the greatest combat and adventure photographer in history.^[4]

RDF

Robert Capa



Gerda Taro



Has spouse



RDF example

01

RDF

URI

http://dbpedia.org/resource/Robert_Capa



URI

<http://example.org/hasSpouse>

URI

http://dbpedia.org/resource/Gerda_Taro



RDF example

01

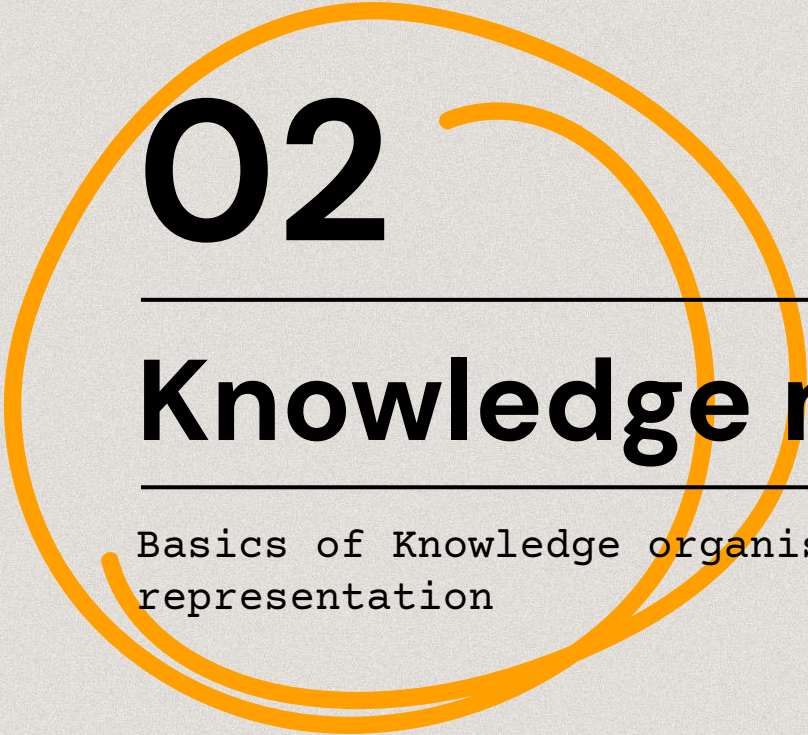
Data

```
<http://dbpedia.org/resource/Robert_Capa>  
<http://example.org/hasSpouse>  
<http://dbpedia.org/resource/Gerda_Taro> .
```

Save as

data.rdf

RDF



02

Knowledge management

Basics of Knowledge organisation and
representation

Classify entities and properties

Ontologies

Person

Entity of the **class**
Person



property

A property relating
objects, e.g. people

Person

Entity of the **class**
Person



Vocabulary terms

Ontologies

Person

<http://dbpedia.org/ontology/Person>



property

<http://example.org/ontology/hasSpouse>

Person

<http://dbpedia.org/ontology/Person>



Namespaces

Ontologies

Person

`http://dbpedia.org/ontology/Person`



property

`http://example.org/ontology/hasSpouse`

Person

`http://dbpedia.org/ontology/Person`



Ontology example

02

Ontologies

Data

```
<http://dbpedia.org/ontology/Person>  
<http://www.w3.org/1999/02/22-rdf-syntax-ns#type>  
<http://www.w3.org/2002/07/owl#Class> .
```

```
<http://dbpedia.org/ontology/Person>  
<http://www.w3.org/2000/01/rdf-schema#label> "Person" .
```

```
<http://dbpedia.org/ontology/Person>  
<http://www.w3.org/2000/01/rdf-schema#comment> "A human being" .
```

```
<http://example.org/ontology/hasSpouse>  
<http://www.w3.org/1999/02/22-rdf-syntax-ns#type>  
<http://www.w3.org/2002/07/owl#ObjectProperty> .
```

```
<http://example.org/ontology/hasSpouse>  
<http://www.w3.org/2000/01/rdf-schema#label> "has spouse".
```

Save as

ontology.rdf or
ontology.owl

RDF is not a format

Serialisations

It's a framework

Conceptual model to formalise the **logical** structure of data in Graphs (VS tables, hierarchies)

Has n serialisations

While the triplet pattern is always respected, there exist several **syntaxes** to serialise RDF.

Has n formats

A **parser** can read and interpret the same information even if served according to different syntaxes and stored in different file **formats**.

RDF serializations

Serialisations

data.nt – ntriples

```
<http://dbpedia.org/resource/Robert_Capa>  
<http://example.org/hasSpouse>  
<http://dbpedia.org/resource/Gerda_Taro> .
```

data.ttl – turtle

```
@prefix db: <http://dbpedia.org/resource/> .  
@prefix ex: <http://example.org/> .  
db:Robert_Capa ex:hasSpouse db:Gerda_Taro .
```

data.xml – XML

```
<?xml version="1.0" encoding="utf-8" ?>  
<rdf:RDF xmlns:db="http://dbpedia.org/resource/" xmlns:ex="http://example.org/">  
  <rdf:Description rdf:about="http://dbpedia.org/resource/Robert_Capa">  
    <ex:hasSpouse rdf:resource="http://dbpedia.org/resource/Gerda_Taro" />  
  </rdf:Description>  
</rdf:RDF>
```

What if triples are not enough

https://en.wikipedia.org/wiki/Robert_Capa

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The Free Encyclopedia

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Robert Capa

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Robert Capa (born **Endre Ernő Friedmann**;^[1] October 22, 1913 – May 25, 1954) was a Hungarian-American war photographer and photojournalist as well as the companion and professional partner of photographer Gerda Taro. He is considered by some to be the greatest combat and adventure photographer in history.^[4]

Graphs

Robert Capa



Has source

Gerda Taro



Has spouse

Named graphs

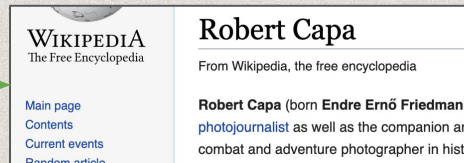
We wrap triples in a graph, a container of triples. We can assign a URI to the **graph** and use the latter as subject of further triples.

Graphs

Robert Capa has
spouse Gerda Taro



Has source



`http://example.org/resource
/Robert_Capa_hasSpouse_Gerd
a_Taro`

`http://example.org/hasSource`

`https://en.wikipedia.org/wiki
/Robert_Capa`

Named graphs serializations

Serialisations

data.nq – nQuads

```
<http://dbpedia.org/resource/Robert_Capa>  
<http://example.org/hasSpouse>  
<http://dbpedia.org/resource/Gerda_Taro>  
<http://example.org/resource/Robert_Capa_hasSpouse_Gerda_Taro> .  
  
<http://example.org/resource/Robert_Capa_hasSpouse_Gerda_Taro>  
<http://example.org/hasSource>  
<https://en.wikipedia.org/wiki/Robert_Capa>  
<http://example.org/resource/Robert_Capa_hasSpouse_Gerda_Taro> .
```

Named graphs serializations

Serialisations

data.trig – Trig

```
@prefix db: <http://dbpedia.org/resource/> .
@prefix ex: <http://example.org/> .
@prefix wiki: <https://en.wikipedia.org/wiki/> .

ex:Robert_Capa_hasSpouse_Gerda_Taro {
  db:Robert_Capa ex:hasSpouse db:Gerda_Taro .
  ex:Robert_Capa_hasSpouse_Gerda_Taro ex:hasSource wiki:Robert_Capa .
}
```



03

Case study

ARTchives

Scope and content

ARTchives

Art historians

The creators of archival collections.

Focus on historians of **Italian Modern Art** (15-16th centuries).

Collections

Materials produced by art historians (**letters, photos, etc.**)

<30 collections.

Keepers

Cultural institutions (archives) **preserving** collections.

6 institutes promoting the project.

Types of entities

ARTchives

Actors

Artists and artistic movements studied by the art historian.

Other art historians.

Universities and research centres.

Contents

Types of materials.
Dates.

Contents: Artists and movements.

Correspondants.

Places. **Bibliography**.

Descriptions
(biographies, scope and content)

Other

Cities and contact addresses.

Vocabularies and data

ARTchives

Wikidata vocab.

Whenever applicable classes and properties are taken from Wikidata.

Wikidata entities

Likewise, entities reuse Wikidata URIs when existing, otherwise new URIs are minted.

Named graphs

All information about a collection, its creator, and the keeper, are stored in a named graph.

Aby Warburg's collection <<https://w3id.org/artchives/collectionfondo-aby-warburg>>

Main subject <<http://www.wikidata.org/prop/direct/P921>>

Federico Zeri <<http://www.wikidata.org/entity/Q1089074>>

The graph of A.W.'s collection <<https://w3id.org/artchives/1598630286-3009102/>> .

Some problems

ARTchives

User data

Duplication of data
(e.g. **labels**).

Wrong
reconciliation.

Misspelling.

Design choices

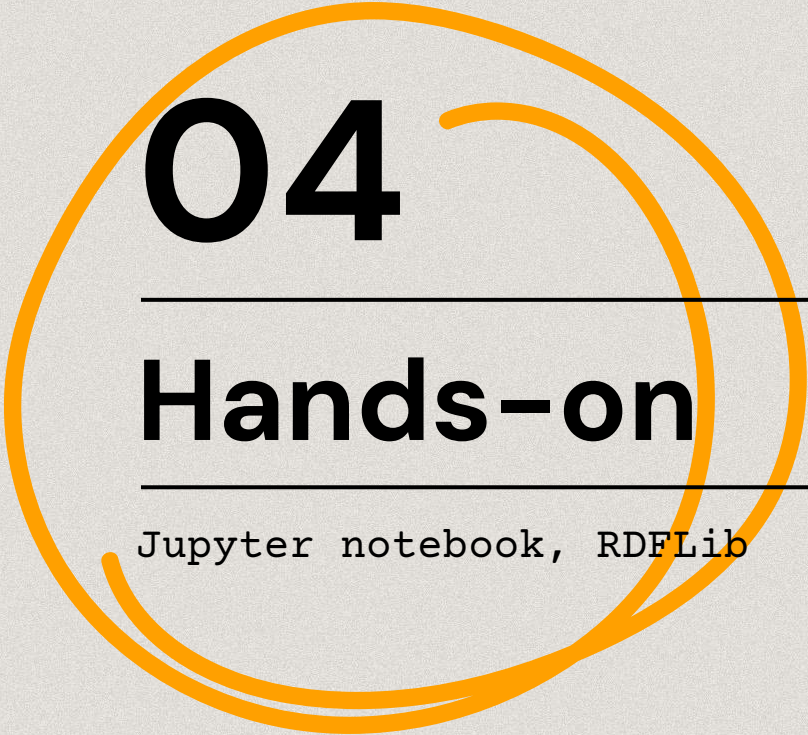
Data about historians
and keepers appear
only in the **graph** of
the first created
collection

(if there are two
collections for the
same historian, only
one includes data
about the historian).

Mistakes

URIs of Wikidata
properties use the wrong
namespace in the online
version.

The data dump you will
use, has the correct
namespaces.



04

Hands-on

Jupyter notebook, RDFLib

Hands-on

Get all the
materials

Download data

Download the data
(resources/artchives.nq) in
a folder for the exercise

Install packages

In the terminal/shell
(if IDE or Jupyter)

```
pip install rdflib  
pip install pprint
```

Tutorial

Open the tutorial:
in GitHub, Colab
or Jupyter (download)

Practice

Choose your environment:
IDE: create a .py file
Jup: create .ipynb file
Colab: new notebook

Solve the problems (time to code!)

Fill in the form with your answers

<https://forms.gle/uWW3j7sXNZ8WF3B89>

Exercise

Review

Review the tutorial

TODO

Come prepared!
Install the python
library with pip
SPARQLWrapper

Thanks!

Do you have any questions?

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https://github.com/marilenadaquino/information_visualization

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